



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Arpit Chakraborty **PRN:25070122047**

Subject: Programming with JAVA **Sem: III** **AY: 2026-27**

Assignment No: 15

TITLE: Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Exercise

1. Create a Login program that throws an exception for invalid password and uses finally block.

```
import java.util.Scanner;
```

```
class BankLogin {
```

```
    static void checkLogin(String password) throws Exception { if
```

```
        (!password.equals("bank@123")) {
```

```
            throw new Exception("Wrong password entered!");
```

```
        }
```

```
        System.out.println("Login successful. Welcome to Online Banking!");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your banking password: "); String
        password = sc.nextLine();

        try {
            checkLogin(password);
        }
        catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
        finally {
            System.out.println("Banking login process completed.");
        }

        sc.close();
    }
}
```

OUTPUT:

```
Output

Enter your banking password: abcd
ERROR!
Error: Wrong password entered!
```

Output

```
Enter your banking password: bank@123
Login successful. Welcome to Online Banking!
Banking login process completed.
```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```
import java.util.Scanner;
```

```
class ATMSystem {
```

```
    static void checkPin(int pin) throws Exception {
```

```
        if (pin != 5678) {
```

```
            throw new Exception("Incorrect ATM PIN!");
```

```
        }
```

```
        System.out.println("PIN accepted. You can continue with your transaction.");
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter your 4-digit PIN: ");
```

```
        int pin = sc.nextInt();
```

```
        try {
```

```
            checkPin(pin);
```

```
        }
```

```
        catch (Exception e) {
```

```
        System.out.println("Error: " + e.getMessage());
    }
    finally {
        System.out.println("ATM PIN verification completed.");
    }

    sc.close();
}
}
```

OUTPUT:

Output

```
Enter your 4-digit PIN: 1234
ERROR!
Error: Incorrect ATM PIN!
ATM PIN verification completed.
```

Output

```
Enter your 4-digit PIN: 5678
PIN accepted. You can continue with your transaction.
ATM PIN verification completed.
```

CONCLUSION:

The Java programs successfully demonstrated **exception handling using throw, throws, and finally**. The throw keyword was used to generate exceptions for invalid passwords and PINs, while throws declared the exceptions in methods. The finally block ensured that the verification process completion message was displayed regardless of whether an exception occurred. Thus, exception handling helps make Java programs more reliable and prevents unexpected program termination.